🌐 Browser Monitors

> **Series:** SYNTH | **Notebook:** 2 of 6 | **Created:** December 2025 | **Last Updated:** 04/04/2026

Creating and Optimizing Browser-Based Synthetic Tests

This notebook covers browser monitors in Dynatrace, including single-URL monitors, browser clickpaths, and performance analysis using the latest Dynatrace platform capabilities.

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Prerequisites

- ✅ Access to a Dynatrace environment with Synthetic Monitoring
- ✅ Completed SYNTH-01 Fundamentals
- ✅ Web application URL to monitor

1. Browser Monitor Types

Single-URL Browser Monitor

Loads a single page and captures performance metrics:

Feature	Description
Execution	Full Chrome browser render
Metrics	W3C Navigation Timing, resource timing
Screenshots	Automatic capture on completion/failure
Validation	Content validation, element checks

Best For:
- Homepage availability
- Landing page performance
- Single page applications (SPA) initial load

Browser Clickpath Monitor

Multi-step user journey simulation:

Feature	Description
Steps	Multiple pages/actions in sequence
Interactions	Click, type, select, wait
State	Cookies maintained within execution (session NOT maintained between executions)
Content Validation	Validate text/elements exist on page

> **Note:** Unlike RUM session properties, browser clickpaths do not support extracting data into variables for use across executions. Content validation allows you to verify expected text/elements exist, but you cannot capture dynamic values for reuse.

Best For:
- Login flows
- Checkout processes
- Form submissions
- Multi-page workflows

Configuration Path

Dynatrace menu → Synthetic → Create synthetic monitor → Create browser monitor

2. Single-URL Monitors

Creating a Single-URL Monitor

1. **URL Configuration**
   - Enter the full URL (https://...)
   - Set viewport size (desktop, tablet, mobile)
   - Configure user agent string

2. **Execution Settings**
   - Frequency: 5-60 minutes
   - Locations: Select public or private
   - Timeout: Maximum execution time

3. **Validation Rules**
   - HTTP status code validation
   - Content validation (text, regex)
   - Element presence checks

Viewport Presets

Preset	Dimensions	Use Case
Desktop	1920x1080	Standard desktop
Laptop	1366x768	Common laptop
Tablet	768x1024	iPad portrait

Preset	Dimensions	Use Case
Mobile	375x667	iPhone 8

```
// List all browser monitors
fetch dt.entity.synthetic_test
| filter isNotNull(browserMonitorSubtype)
| fields id, entity.name, isEnabled, browserMonitorSubtype
| sort entity.name asc
| limit 50
```

```
// Browser monitor execution results (last 24h)
fetch bizevents, from: now() - 24h
| filter event.provider == "dynatrace.synthetic"
| filter matchesValue(event_type, "*browser*")
| fields timestamp,
         monitor = synthetic_test_id,
         location = synthetic_location_id,
         availability = execution_success,
         response_time_ms = toDouble(total_duration)
| sort timestamp desc
| limit 100
```

3. Browser Clickpaths

Creating Clickpath Monitors

Option 1: Record with Browser Extension

1. Install Dynatrace Synthetic Recorder (Chrome extension)
2. Start recording session
3. Perform user journey in browser
4. Stop recording and export to Dynatrace

Option 2: Manual Script Creation

Define steps programmatically using the script editor.

Common Actions

Action	Description	Example
`navigate`	Go to URL	Navigate to login page
`click`	Click element	Click login button
`type`	Enter text	Type username
`selectOption`	Select dropdown	Select country
`wait`	Wait for condition	Wait for element visible
`javascript`	Execute JS	Custom validation

Element Selectors

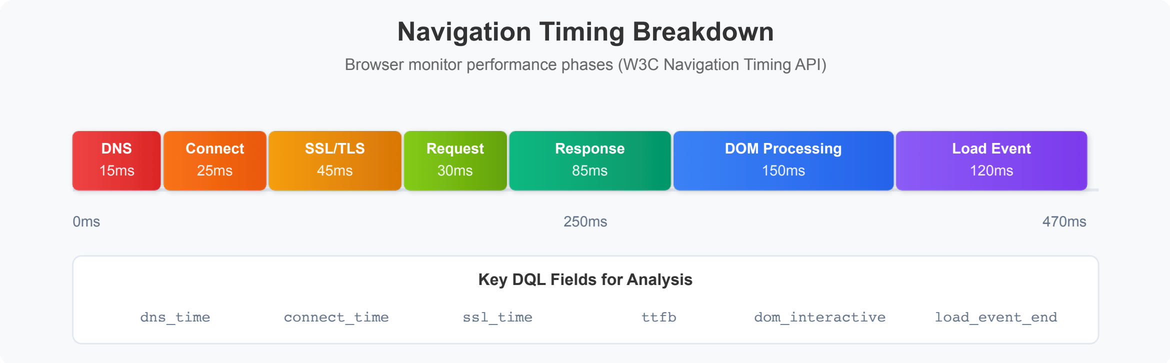
Selector Type	Example	Best Practice
CSS	`#login-btn`	Preferred - stable
XPath	`//button[@id='login']`	Complex structures
Link Text	`Login`	Simple links
Data Attribute	`[data-testid='login']`	Test automation

```
// Clickpath step performance analysis
fetch bizevents, from: now() - 24h
| filter event.provider == "dynatrace.synthetic"
| filter isNotNull(step_title)
| summarize {
    avg_duration_ms = avg(toDouble(step_duration)),
    max_duration_ms = max(toDouble(step_duration)),
    executions = count()
  }, by: {synthetic_test_id, step_title}
| sort avg_duration_ms desc
| limit 30
```

4. Performance Metrics

W3C Navigation Timing

Browser monitors capture detailed timing metrics based on the W3C Navigation Timing API:



Key Performance Metrics

Metric	Description	Good	Needs Work
First Contentful Paint	First content rendered	< 1.8s	> 3.0s
Largest Contentful Paint	Largest element rendered	< 2.5s	> 4.0s
Time to Interactive	Page fully interactive	< 3.8s	> 7.3s
Total Blocking Time	Main thread blocked	< 200ms	> 600ms
Cumulative Layout Shift	Visual stability	< 0.1	> 0.25

> **Note:** Targets based on Google's Core Web Vitals. Actual acceptable thresholds vary by application type and user expectations. Set baselines based on your own monitoring data.

```
// Browser monitor performance breakdown
fetch bizevents, from: now() - 24h
| filter event.provider == "dynatrace.synthetic"
| filter matchesValue(event_type, "*browser*")
| summarize {
    avg_dns_ms = avg(toDouble(dns_lookup_time)),
    avg_connect_ms = avg(toDouble(tcp_connect_time)),
    avg_ssl_ms = avg(toDouble(ssl_handshake_time)),
    avg_ttfb_ms = avg(toDouble(time_to_first_byte)),
    avg_dom_interactive_ms = avg(toDouble(dom_interactive_time)),
    avg_load_ms = avg(toDouble(total_duration)),
    executions = count()
  }, by: {synthetic_test_id}
| sort avg_load_ms desc
| limit 20
```

```
// Performance trend over time
fetch bizevents, from: now() - 7d
| filter event.provider == "dynatrace.synthetic"
| filter matchesValue(event_type, "*browser*")
| makeTimeseries {
    avg_response_time = avg(toDouble(total_duration)),
    p95_response_time = percentile(toDouble(total_duration), 95)
  }, interval: 1h
```

5. Validation and Assertions

Content Validation

Validation Type	Description	Example
Text Present	Page contains text	"Welcome"
Text Absent	Page doesn't contain	"Error"
Regex Match	Pattern matching	`Order #\d{6}`
Element Exists	DOM element present	`#success-message`
Element Content	Element has text	Button says "Submit"

HTTP Validation

Check	Default Behavior	Customization
Status Code	Fails on 4xx/5xx (400-599)	Can configure to ignore specific codes
Response Body	Content validation	Text/regex matching
Screenshots	Captured on success/failure	Automatic

> **Note:** Response size validation and header validation are not available as built-in options. Use content validation or JavaScript steps for advanced checks.

Visual Validation

- Automatic screenshots on success/failure
- Visual comparison (pixel diff)
- Layout validation

```
// Failed browser monitor executions
fetch bizevents, from: now() - 24h
| filter event.provider == "dynatrace.synthetic"
| filter matchesValue(event_type, "*browser*")
| filter execution_success == false
| fields timestamp,
         monitor = synthetic_test_id,
         location = synthetic_location_id,
         error = error_message
| sort timestamp desc
| limit 50
```

6. Analyzing Browser Results

```
// Browser monitor availability by location
fetch bizevents, from: now() - 24h
| filter event.provider == "dynatrace.synthetic"
| filter matchesValue(event_type, "*browser*")
| summarize {
    total = count(),
    successful = countIf(execution_success == true),
    failed = countIf(execution_success == false)
  }, by: {synthetic_test_id, synthetic_location_id}
| fieldsAdd availability_pct = round((successful * 100.0) / total, decimals: 2)
| sort availability_pct asc
| limit 30
```

```
// Response time distribution by location
fetch bizevents, from: now() - 24h
| filter event.provider == "dynatrace.synthetic"
| filter matchesValue(event_type, "*browser*")
| filter execution_success == true
| summarize {
    min_ms = min(toDouble(total_duration)),
    avg_ms = avg(toDouble(total_duration)),
    p50_ms = percentile(toDouble(total_duration), 50),
    p95_ms = percentile(toDouble(total_duration), 95),
    max_ms = max(toDouble(total_duration)),
    executions = count()
  }, by: {synthetic_location_id}
| sort avg_ms desc
| limit 20
```

```
// Slowest page loads (outliers)
fetch bizevents, from: now() - 24h
| filter event.provider == "dynatrace.synthetic"
| filter matchesValue(event_type, "*browser*")
| filter execution_success == true
| filter toDouble(total_duration) &gt; 5000  // &gt; 5 seconds
| fields timestamp,
         monitor = synthetic_test_id,
         location = synthetic_location_id,
         response_time_ms = toDouble(total_duration)
| sort response_time_ms desc
| limit 20
```

Summary

In this notebook, you learned:

✅ **Browser monitor types** - Single-URL vs clickpath monitors
✅ **Creating monitors** - URL configuration, viewports, locations
✅ **Clickpath automation** - Recording, actions, selectors
✅ **Performance metrics** - W3C timing, Core Web Vitals
✅ **Validation** - Content, HTTP, and visual checks
✅ **Analysis queries** - Availability, response times, failures

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Next Steps

Continue to **SYNTH-03: HTTP Monitors** to learn about lightweight API monitoring.

References

- Browser Monitors
- Browser Clickpaths
- Synthetic Recorder
- Performance Metrics

This notebook was AI-generated from community-submitted and publicly available sources. This notebook series is not officially supported by Dynatrace. Always verify information against official Dynatrace documentation.